

- (c) The class `Fence` stores data about the fences. Each fence has a height in cm and a risk number.

The risk is a whole number between 1 and 5 inclusive. A risk of 1 means the fence is the easiest type to jump. A risk of 5 means the fence is the hardest type to jump.

Fence	
Height : INTEGER	stores the height of the fence in cm the height is between 70 and 180 inclusive
Risk : INTEGER	stores the risk as a whole number between 1 and 5 inclusive
Constructor()	initialises Height and Risk to its parameter values
GetHeight()	returns the height of the fence
GetRisk()	returns the risk of the fence

- (i) Write program code to declare the class `Fence`, its constructor and get methods.

Use your programming language's appropriate constructor.

All attributes must be private.

If you are writing in Python, include attribute declarations using comments.

Save your program.

Copy and paste the program code into part **2(c)(i)** in the evidence document.

[4]

- (ii) The array `Course` stores **four** `Fence` objects. The user inputs the height and risk for each fence, and these are validated before each fence is created.

Amend the main program to:

- declare the local array `Course`
- take as input the data for **four** fences from the user
- loop the input until both the height and risk are valid for each fence
- create an instance of `Fence` for each of the **four** valid fences and store each instance in the array.

Save your program.

Copy and paste the program code into part **2(c)(ii)** in the evidence document.

[5]